

Experiment 'p_2_4_c_product' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_4_c_product

Natural language statement: Write a method that takes two integers and then returns the product.

Method signature: p_2_4_c_product(x: int, y: int) returns (result: int)

Ensures

- `result == x * y`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes two integers and then returns the product.

The signature should be:

`p_2_4_c_product(x: int, y: int) returns (result: int)`

The method should respect the following contract:

`ensures result == x * y`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdXArU2fyAE9SGZRTdYkcmF6PW11

Model: gpt-5.5-2026-04-23

Created at: 1780484084

Finish reason: stop

Usage: 260 tokens in, and 22 tokens out

```
{  
  result := x * y;  
}
```

Final Program

```
method p_2_4_c_product(x: int, y: int) returns (result: int)  
  ensures result == x * y  
{  
  result := x * y;  
}
```

Total Token Usage

Input tokens: 260

Output tokens: 22

Reasoning tokens: 0

Sum of ‘total tokens’: 282

Experiment Timings

Overall Experiment started at 1780484083808, ended at 1780484089187, lasting 5379ms (5.38 seconds)

Iteration #1 started at 1780484083810, ended at 1780484089187, lasting 5377ms (5.38 seconds)